

Day 34 - Standard ACLs

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10:46 AM

ACL Information

What Are Access Control Lists (ACLs)?

Access Control Lists (ACLs) are traffic filters which contain ordered lists of rules called 'Access Control Entries' (ACEs).

- **ACLs** function as a packet filter, as they instruct the router to **permit** or **deny** traffic based on the defined **ACEs**.
- **ACLs** can filter traffic based on **source/destination** IP addresses, **source/destination** Layer 4 ports, and etc.

ACL Config General Info

- **ACLs** are configured **globally** on the router (**global configuration mode**), but they must be **applied** onto an interface in order to take effect.
- **ACLs** are applied either **inbound** or **outbound** on interfaces.
- When the router checks a packet against the configured **ACL**, it processes the **ACEs** in order, from top to bottom.
- If the packet matches one of the **ACEs** in the **ACL**, the router will take action (apply the **ACE** rule) and stops processing the **ACL** (all entries below the matching entry will be ignored).

Implicit Deny

Implicit Deny is a feature that is present in all ACLs. The **implicit deny** tells the router to **deny** all traffic that doesn't match any of the configured ACEs in the ACL.

ACL Types

There are two types of ACLs with both containing two subtypes:

1. Standard ACLs

- **Standard ACLs** match based on **source IP address** only
- **Standard ACLs** can use the number range (**1 - 99**) and (**1300 - 1999**)
- They should be applied as close to the destination as possible.
- They include two subtypes:
 1. Standard **numbered** ACLs
 2. Standard **named** ACLs

2. Extended ACLs

- **Extended ACLs** match based on **source/destination IP address, source/destination port address**, etc.
- **Extended ACLs** can use the number range (**100 - 199**) and (**2000 - 2699**)
- They should be applied as close to the source as possible.
- They include two subtypes:
 1. Extended **numbered** ACLs
 2. Extended **named** ACLs

Numbered ACLs

- Different types of **numbered ACLs** have a different range of numbers that they can use.
- So, each range can be used to identify what type of **numbered ACL** we're looking it.

Standard ACL Types

1. Standard Numbered ACLs

Standard **numbered** ACLs are one of the Standard ACL subtypes. They are identified with a number.

- (Ex. **ACL 1, ACL 2**)

2. Standard Named ACLs

Standard **named** ACLs are one of the Standard ACL subtypes. They are identified with a name.

- (Ex. **FIRST_ACL**, **CCNA_ACL**)

Standard ACL Configuration

There are two ways to create a standard ACLs:

1. Global configuration / directly

- This method only allows you to create **numbered ACLs**
- Done through the command:
 - `access-list STANDARD-ACL-NUMBER { deny | permit } IP WILDCARD-MASK`
- **Examples:**
 - `access-list 1 deny 1.1.1.1 0.0.0.0`
 - `access-list 1 deny 1.1.1.1 <--` (*only possible in Standard ACLs*)
 - `access-list 1 deny host 1.1.1.1`
 - `access-list 1 permit any`
 - `access-list 1 permit 0.0.0.0 255.255.255.255`

2. ACL configuration mode

- Done through the command:
 - `ip access-list standard NAME/NUMBER`
- Which puts you in **Standard ACL configuration** mode. **ACEs** can be set through the following command format:
 - `[entry-number] { deny | permit } ip wildcard-mask`

Access-list arguments

After the '**deny** | **permit**' part of your command, you usually specify the **ip** and **wildcard-mask**. But, there are some arguments you could use as well:

1. host

- Instead of specifying a **wildcard-mask** for a host address (**/32**) like this:
 - `access-list 1 deny 1.1.1.1 0.0.0.0`
- You can use the **host** argument and the router will automatically translate it into a host address (**/32**):
 - `access-list 1 deny host 1.1.1.1`

1. any

- Instead of specifying an **ip** and **wildcard-mask** to match/deny all traffic like this:
 - `access-list 1 permit 0.0.0.0 255.255.255.255`
- You could use the **any** argument, which will match/deny all traffic:
 - `access-list 1 permit any`

Notes on how routers handle ACLs

- The router may re-order the **/32** entries.
- This improves the efficiency of processing the **ACL**.
- It does not change the effect of the **ACL**.
- This applies to both **standard named** and **standard numbered ACLs**.